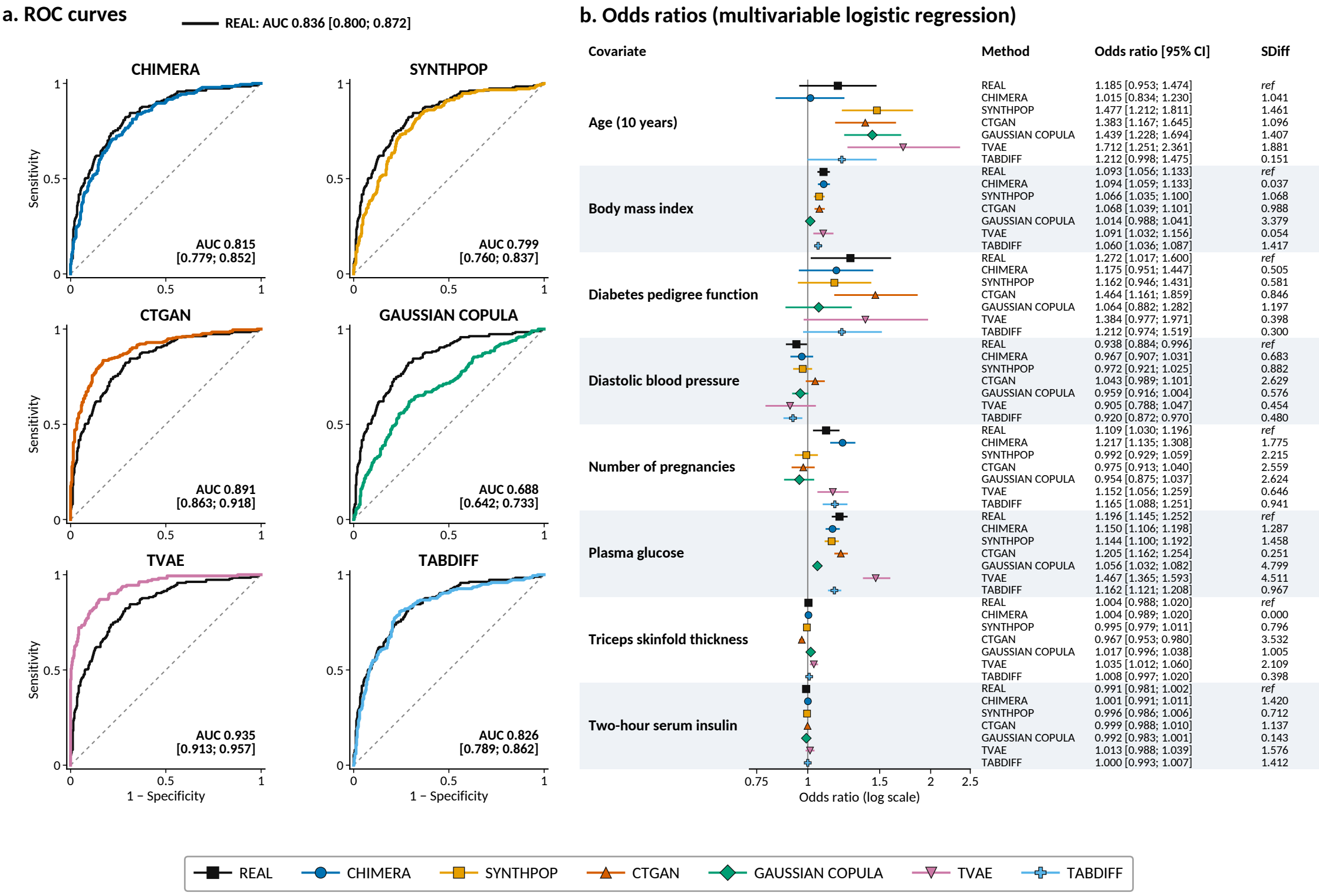

a. ROC curves
REAL: AUC 0.836 [0.800; 0.872]
CHIMERA
AUC 0.815 [0.779; 0.852]
SYNTHPOP
AUC 0.799 [0.760; 0.837]
CTGAN
AUC 0.891 [0.863; 0.918]
GAUSSIAN COPULA
AUC 0.688 [0.642; 0.733]
TVAE
AUC 0.935 [0.913; 0.957]
TABDIFF
AUC 0.826 [0.789; 0.862]
Sensitivity
1 – Specificity
b. Odds ratios (multivariable logistic regression)
Covariate
Method
Odds ratio [95% CI]
SDiff
Age (10 years)
REAL 1.185 [0.953; 1.474] ref
CHIMERA 1.015 [0.834; 1.230] 1.041
SYNTHPOP 1.477 [1.212; 1.811] 1.461
CTGAN 1.383 [1.167; 1.645] 1.096
GAUSSIAN COPULA 1.439 [1.228; 1.694] 1.407
TVAE 1.712 [1.251; 2.361] 1.881
TABDIFF 1.212 [0.998; 1.475] 0.151
Body mass index
REAL 1.093 [1.056; 1.133] ref
CHIMERA 1.094 [1.059; 1.133] 0.037
SYNTHPOP 1.066 [1.035; 1.100] 1.068
CTGAN 1.068 [1.039; 1.101] 0.988
GAUSSIAN COPULA 1.014 [0.988; 1.041] 3.379
TVAE 1.091 [1.032; 1.156] 0.054
TABDIFF 1.060 [1.036; 1.087] 1.417
Diabetes pedigree function
REAL 1.272 [1.017; 1.600] ref
CHIMERA 1.175 [0.951; 1.447] 0.505
SYNTHPOP 1.162 [0.946; 1.431] 0.581
CTGAN 1.464 [1.161; 1.859] 0.846
GAUSSIAN COPULA 1.064 [0.882; 1.282] 1.197
TVAE 1.384 [0.977; 1.971] 0.398
TABDIFF 1.212 [0.974; 1.519] 0.300
Diastolic blood pressure
REAL 0.938 [0.884; 0.996] ref
CHIMERA 0.967 [0.907; 1.031] 0.683
SYNTHPOP 0.972 [0.921; 1.025] 0.882
CTGAN 1.043 [0.989; 1.101] 2.629
GAUSSIAN COPULA 0.959 [0.916; 1.004] 0.576
TVAE 0.905 [0.788; 1.047] 0.454
TABDIFF 0.920 [0.872; 0.970] 0.480
Number of pregnancies
REAL 1.109 [1.030; 1.196] ref
CHIMERA 1.217 [1.135; 1.308] 1.775
SYNTHPOP 0.992 [0.929; 1.059] 2.215
CTGAN 0.975 [0.913; 1.040] 2.559
GAUSSIAN COPULA 0.954 [0.875; 1.037] 2.624
TVAE 1.152 [1.056; 1.259] 0.646
TABDIFF 1.165 [1.088; 1.251] 0.941
Plasma glucose
REAL 1.196 [1.145; 1.252] ref
CHIMERA 1.150 [1.106; 1.198] 1.287
SYNTHPOP 1.144 [1.100; 1.192] 1.458
CTGAN 1.205 [1.162; 1.254] 0.251
GAUSSIAN COPULA 1.056 [1.032; 1.082] 4.799
TVAE 1.467 [1.365; 1.593] 4.511
TABDIFF 1.162 [1.121; 1.208] 0.967
Triceps skinfold thickness
REAL 1.004 [0.988; 1.020] ref
CHIMERA 1.004 [0.989; 1.020] 0.000
SYNTHPOP 0.995 [0.979; 1.011] 0.796
CTGAN 0.967 [0.953; 0.980] 3.532
GAUSSIAN COPULA 1.017 [0.996; 1.038] 1.005
TVAE 1.035 [1.012; 1.060] 2.109
TABDIFF 1.008 [0.997; 1.020] 0.398
Two-hour serum insulin
REAL 0.991 [0.981; 1.002] ref
CHIMERA 1.001 [0.991; 1.011] 1.420
SYNTHPOP 0.996 [0.986; 1.006] 0.712
CTGAN 0.999 [0.988; 1.010] 1.137
GAUSSIAN COPULA 0.992 [0.983; 1.001] 0.143
TVAE 1.013 [0.988; 1.039] 1.576
TABDIFF 1.000 [0.993; 1.007] 1.412
0.75
1
1.5
2
2.5
Odds ratio (log scale)
REAL
CHIMERA
SYNTHPOP
CTGAN
GAUSSIAN COPULA
TVAE
TABDIFF